Real collisions of zeros under the Gaussian backward heat flow

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Abstract

Csordas and Escassut asked whether strict positivity of the first Laguerre expression at every positive time of the backward heat flow forces a strip entire function of order less than two to have only real zeros. We prove their conjecture. If an initial zero is nonreal, a real multiple zero occurs at a finite point and at a positive time no later than half the square of the initial strip half-width. The time bound is sharp. The main issue is ruling out loss of collisions at horizontal infinity. We count the upper zeros of canonical approximants with a rational weight whose pair interactions cancel in a companion sum of imaginary parts. Applying the balance at two spatial scales controls the tails of the collision measures. The proof also gives a multiplicity count, a version for the full canonical strip class on its natural time interval, and a weak Laguerre test that detects collisions of either parity. For the Riemann heat flow the weak test characterizes the Newman constant. Its nonnegativity at all positive times remains to be proved.

1 A Gaussian converse and a collision count

Let $D = d/dz$ and $L(g) = (g')^2 - gg''$. The strip class $\mathcal{S}(A)$ of Csordas and Escassut consists of nonzero real entire functions

$$f(z) = e^{-az^2} g(z), \qquad g(z) = Cz^d e^{bz} \prod_{\rho \neq 0} (1 - z/\rho) e^{z/\rho}, \tag{1.1}$$

where $a \geq 0$, $b, C \in \mathbb{R}$, $C \neq 0$, d is a nonnegative integer, $|\operatorname{Im} \rho| \leq A < \infty$, and $\sum_{\rho \neq 0} |\rho|^{-2} < \infty$. All zeros are counted with multiplicity. Within this class, f belongs to the Laguerre–Pólya class precisely when its zeros are all real. At a real multiple zero the Laguerre expression vanishes.

Theorem 1.1 (Csordas–Escassut Gaussian conjecture)**.** *Let $f \in \mathcal{S}(A)$ have order less than two and put $F(t, z) = e^{-tD^2} f(z)$. If $L(F(t, \cdot))(x) > 0$ for every $t > 0$ and $x \in \mathbb{R}$, all zeros of f are real.*

The source is the conjecture accompanying equation (3.3) of [1, pp. 336–337]: for their Gaussian symbol $\varphi(z) = e^{-z^2}$ one sets $t = \mu^2$. Their Open Problem 3.3 asks for a classification of symbols of order below two. The same paper's Conjecture 3.5 concerns the cosine symbol. The Gaussian implication follows from the geometric statement below.

Theorem 1.2 (Finite real collisions)**.** *Suppose f has the representation (1.1), and put $F(t, z) = e^{-tD^2} f(z)$. Take $0 \leq t < 1/(4a)$, with no upper restriction if $a = 0$. If f has a nonreal zero, then for some real x_* and some t_*,*

$$0 < t_* \leq \frac{A^2}{2 + 4aA^2}, \qquad F(t_*, x_*) = F_z(t_*, x_*) = 0.$$

More precisely, if M is the number of initial upper-half-plane zeros, then in the extended nonnegative integers

$$M = \sum_{(t,x)} \left\lfloor \frac{m(t, x)}{2} \right\rfloor. \tag{1.2}$$

The sum is over positive times in the indicated domain and real multiple zeros, and $m(t,x)$ is their spatial multiplicity.

For $f(z) = e^{-az^2}(z^2 + A^2)$ with $A > 0$, the bound for the first collision is attained. For polynomial initial data with $a = 0$ the count follows by tracking finitely many roots. Strip contraction puts all zeros on the real line by time $A^2/2$ [2, 3]. For infinite products the corresponding finite-dimensional statement alone gives no control over where the approximating polynomials collide. We obtain that control by comparing weighted counts at two different scales. No separation, density assumption, or numerical verification of individual zeros enters the argument.

2 Canonical truncation and the range of the weights

First let $a = 0$. Keep the nonzero zeros in increasing disks, retaining conjugate pairs, and let $f_n = e^{b_n z} P_n(z)$ be the resulting canonical truncations, with b_n real and P_n a real polynomial. They converge to f locally uniformly. For each $\delta > 0$ there is C_δ independent of n such that

$$|f_n(z)| \le C_\delta \exp(\delta |z|^2).$$

Indeed $\log|(1-u)e^u| \le C|u|^2$ globally. A finite initial group of factors has logarithmic maximum $O(1 + |z| + \log(2 + |z|))$, and the remaining reciprocal-square sum can be made arbitrarily small. The bound permits canonical products of order two and minimal type.

We use the Gaussian integral

$$F_n(t,z) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-v^2} f_n(z + 2i\sqrt{t}\, v)\, dv \tag{2.1}$$

and the same formula for F. On a compact set of complex t and z the preceding bound with sufficiently small δ supplies an integrable majorant. The integral is even in $\sqrt{t}$, so is entire in t, and Taylor expansion gives $F_t = -F_{zz}$ (subscripts in this identity denote partial derivatives). On each $0 \le t \le T$ the transforms and their spatial derivatives converge locally uniformly in z, uniformly in t. Also

$$\sup_{n,\, 0\le t\le T} \log^+ \max_{|z|\le R} |F_n(t,z)| = o(R^2). \tag{2.2}$$

For this last estimate bound $|z + 2i\sqrt{t}\, v|^2$ by $2|z|^2 + 8Tv^2$ and take δ arbitrarily small. The transform is injective on this growth class. Following (2.1) by the real Gaussian displacement of the same variance amounts to averaging $f(z + 2\sqrt{t}\,(u + iv))$ against $e^{-u^2-v^2}/\pi$. Angular averaging gives $f(z)$, with Fubini justified by the same minimal growth. Hence $F(t,\cdot) \not\equiv 0$.

For a fixed approximant,

$$F_n(t,z) = e^{b_n z - t b_n^2}\, (e^{-tD^2} P_n)(z - 2tb_n).$$

The polynomial form of the Gaussian strip theorem gives $|\operatorname{Im} z| \le \sqrt{(A^2 - 2t)_+}$ for its zeros [2, 3]. Hurwitz's theorem gives the same conclusion for F. Put $\ell = A + 2$, $w_0 = i\ell$. On $[0,T]$, $F_n(t, w_0)$ are uniformly bounded away from zero and $|F_n'/F_n|(t, w_0)$ are uniformly bounded. This follows by local convergence, compactness and nonvanishing at w_0 for each of the finitely many discarded initial approximants. Primes on spatial slices henceforth denote spatial derivatives.

Write $z_k = x_k + iy_k$ for the zeros of the current approximant at time t. For $L \ge 4\ell$ put $d_k = (L^2 + x_k^2)^{1/2}$, $h_k = L^2/d_k^2$. An important uniform bound is

$$\epsilon_L := \sup_{n,\, 0\le t\le T} \sum_k d_k^{-2} \longrightarrow 0. \tag{2.3}$$

Along the imaginary axis the real exponential multiplier has constant modulus. Consequently

$$\log \frac{|F_n(t, iL)|}{|F_n(t, i\ell)|} = \frac{1}{2} \sum_k \log\left(1 + \frac{(L-\ell)(L+\ell-2y_k)}{x_k^2 + (\ell - y_k)^2}\right).$$

Each summand is at least cL^2/d_k^2 after allowing for the factor $1/2$, by $\log(1+v) \geq v/(1+v)$ for $v \geq 0$ and $|y_k| \leq A$. Now (2.3) follows from (2.2) and the uniform lower bound at $i\ell$.

Define

$$g_L(z) = \frac{L^2}{L^2+z^2}, \qquad r_L(z) = \frac{z}{L^2+z^2}, \qquad B = b_n - \sum_k r_L(z_k) = \operatorname{Re} \frac{F_n'(t,iL)}{F_n(t,iL)}.$$

The latter identity follows by averaging the logarithmic derivatives at iL and $-iL$ and using conjugation. Since b_n is real, $-\operatorname{Im}(F_n'/F_n)(t,w_0) \geq 2\sum_k |w_0 - z_k|^{-2}$. Subtract the logarithmic derivatives at iL and w_0 and apply Cauchy–Schwarz, using $|iL - z_k| \asymp d_k$, to obtain

$$|B|/L \leq C/L + C\sqrt{\epsilon_L}, \qquad \delta_L := L^{-2} + \epsilon_L + \sup_{n,t} |B|/L \longrightarrow 0.$$

All ensuing bounds are uniform over truncations and $t \in [0,T]$. Their constants may depend on f, A, T.

3 Weighted zero counts and their variation

Fix $T > A^2/2$ and $0 \leq s < T$. For this section abbreviate g_L, r_L by g, r.

Write $U = \{i : y_i > 0\}$ and let H be the primitive of g in $|\operatorname{Im} z| < L$ with $H(0) = 0$. Define

$$N(t) = \sum_{i \in U} \operatorname{Re} g(z_i), \qquad Y(t) = \sum_{i \in U} \operatorname{Im} H(z_i).$$

On the strip $|y| \leq A$, at $z = x + iy$, with $d = (L^2 + x^2)^{1/2}$ and $h = L^2/d^2$, direct differentiation gives

$$\operatorname{Re} g \asymp h, \quad |g| \leq Ch, \quad |g'| \leq Ch/d, \quad |\operatorname{Im} g| \leq C|y|h/d,$$
$$|r| \leq C/d, \qquad |r'| \leq C/d^2, \qquad |\operatorname{Im} r| \leq C|y|/d^2.$$

For example $\operatorname{Re}(L^2 + z^2) = d^2 - y^2 \geq 3d^2/4$ and $g' = -2zL^2/(L^2+z^2)^2$. The imaginary-part bounds follow by integration from the real axis. Similarly $\operatorname{Im} H(x+iy) = \int_0^y \operatorname{Re} g(x+iv)\, dv \asymp yh$ for $y > 0$. In particular $N \asymp \sum_U h_i$ and $0 \leq Y \leq CAN$.

At a time of simple zeros, the zero evolution equation is $z_i' = 2b_n + 2\sum_{k \neq i} (z_i - z_k)^{-1}$. For $i \in U$ put

$$S_i = \sum_{k \notin U} \frac{1}{|z_i - z_k|^2}, \qquad E_i = \sum_{k \notin U} \left(\frac{1}{z_i - z_k} + r(z_k) \right).$$

Adding $r(z_k)$ to the internal interactions as well leaves the one-body term $B + r(z_i)$. The compensated external terms satisfy

$$|E_i| \leq C d_i \sqrt{\epsilon_L S_i}, \tag{3.1}$$

$$\operatorname{Im} E_i \leq -y_i S_i - \frac{1}{4y_i} + CA\epsilon_L. \tag{3.2}$$

For (3.1), the numerator of a compensated summand is $L^2 + z_i z_k$ and its denominator is $(z_i - z_k)(L^2 + z_k^2)$. The numerator has modulus at most $C d_i d_k$, so Cauchy–Schwarz applies. For (3.2) the imaginary part of the uncompensated sum is $-y_i S_i + \sum_{k \notin U} y_k/|z_i - z_k|^2$; its conjugate partner alone contributes $-1/(4y_i)$ to the second term.

The internal pairs disappear from the derivative of Y by the exact algebraic identity

$$\frac{g(z) - g(w)}{z - w} + g(z)r(w) + g(w)r(z) = 0.$$

For the derivative of N apply $\partial_z + \partial_w$ to this identity. The residual contribution of an internal pair before taking real parts and restoring the factor two is $-g(z)r'(w) - g(w)r'(z)$. Its total absolute value over internal pairs is at most $C\epsilon_L N$.

The one-body term in N' has modulus at most $C(|B|/L + L^{-2})N$. For Y' its imaginary part is bounded above by $CA(|B|/L + L^{-2})N$, using B real and the bounds on $\operatorname{Im} g, \operatorname{Im} r$. Combining these estimates and (3.1) gives

$$|N'| \le C\delta_L N + C\sqrt{\epsilon_L} \sum_{i\in U} h_i \sqrt{S_i}. \tag{3.3}$$

In Y', the contribution $\operatorname{Im}(g_i E_i)$ is $\operatorname{Re} g_i \operatorname{Im} E_i + \operatorname{Im} g_i \operatorname{Re} E_i$. The latter is bounded by $Cy_i h_i \sqrt{\epsilon_L S_i}$. Young's inequality absorbs it in half of the negative $y_i h_i S_i$ term in (3.2), at the cost $Cy_i h_i \epsilon_L$. Since $y_i S_i + 1/y_i \ge 2\sqrt{S_i}$, we obtain

$$Y' \le -c \sum_{i\in U} h_i \sqrt{S_i} + CA\delta_L N \tag{3.4}$$

with $c > 0$ independent of L, n, t.

3.1 Real collisions of the approximants

For a fixed polynomial the exceptional times are finite. Its discriminant as a polynomial in t is nonzero. Indeed, at large t, rescaling the space variable by $\sqrt{t}$ and removing the real shift gives the heat transform of the leading monomial in the limit, a Hermite polynomial with simple roots. At a real zero of multiplicity m at time t_*, rescale $z - x_* = \sqrt{|t - t_*|}\,\xi$ after removing the moving real exponential shift. The leading rescaled polynomial is $e^{-\operatorname{sgn}(t-t_*)D_\xi^2}\xi^m$ up to a nonzero scalar. For $t > t_*$ its roots are all real and simple. For $t < t_*$ they are purely imaginary and simple, with the single zero at the origin present exactly for odd m. Rouché's theorem and reality give the same real/nonreal counts for nearby times. Consequently Y is continuous and N loses exactly $\lfloor m/2 \rfloor g(x_*)$ at a real collision. A nonreal collision causes no jump. Between exceptional times (3.3)–(3.4) hold. Integration to their endpoints is legitimate since polynomial zero branches have convergent local Puiseux expansions. $N(s)$ denotes its right-continuous value even if s is exceptional.

Below we write N_R or N_L when the scale matters and $h_L(x) = L^2/(L^2 + x^2)$. Let $V_L(t) = \int_s^t |N'(\tau)|\, d\tau$, excluding the finitely many jump times. Integrating (3.4), using $Y(t) \ge 0$ and $Y(s) \le CAN(s)$, then using (3.3), yields

$$V_L(t) \le C_A \sqrt{\epsilon_L} N(s) + C_A \delta_L (1 + \sqrt{\epsilon_L}) \int_s^t N(\tau)\, d\tau.$$

All jumps are downward, so $N(t) \le N(s) + V_L(t)$. Grönwall's inequality on the fixed horizon implies $\sup_{[s,T]} N(t) \le C_{A,T} N(s)$ for all sufficiently large L, and then

$$V_L(T) \le \eta_L N(s), \qquad \eta_L = C_{A,T}(\sqrt{\epsilon_L} + \delta_L)(1 + \sqrt{\epsilon_L}) \longrightarrow 0.$$

Let ν_n be the counting measure of collisions on $(s, T]$ projected to the real axis, with weight $\lfloor m/2 \rfloor$ for each real zero of multiplicity m. Since $N(T) = 0$ by strip contraction, we have

$$(1 - \eta_L)N(s) \le \int_{\mathbb{R}} \frac{L^2}{L^2 + x^2}\, d\nu_n(x) \le (1 + \eta_L)N(s). \tag{3.5}$$

4 Passing the collision count to the entire function

Write M for the number of upper zeros of f and choose $T > A^2/2$. Use the balance (3.5) with $s = 0$. Its right-hand initial count takes each retained upper zero z of f with weight $\operatorname{Re} g_L(z)$. Consequently at fixed L it converges as $n \to \infty$ to

$$N_L^f(0) = \sum_{\operatorname{Im} z > 0,\ f(z)=0} \operatorname{Re} g_L(z).$$

This is finite by (1.1) and the comparison $\operatorname{Re} g_L(z) \asymp L^2/(L^2 + (\operatorname{Re} z)^2)$.

Applying the estimate at a second scale gives tightness for the weighted measures. For $R \geq L$ and $|x| \geq R$, $h_L(x)/h_R(x) \leq 2L^2/R^2$, where $h_R(x) = R^2/(R^2 + x^2)$. By (3.5) used at scale R,

$$\int_{|x|\geq R} h_L(x)\, d\nu_n(x) \leq \frac{2L^2}{R^2}(1+\eta_R)N_R(0) \leq CL^2\epsilon_R \;\longrightarrow\; 0 \qquad (R \to \infty)$$

uniformly in n at each fixed large L. This step also covers order two and minimal type.

We describe the local limit of ν_n. If F has a real multiple zero x_* of multiplicity m at time t_*, the heat equation implies that

$$u^{-m}F(t_* + u^2, x_* + uv) \longrightarrow \frac{\partial_z^m F(t_*, x_*)}{m!}\, e^{-D_v^2} v^m \quad (u \to 0)$$

locally uniformly in v. This quotient is analytic in (u, v) at $u = 0$ by the joint Taylor expansion, since the coefficient of $u^{2j+k}v^k$ involves the spatial derivative of order $2j + k$. Simplicity of the Hermite zeros gives m local branches analytic in u. At real times just after t_* all their zeros are real by conjugation. Just before t_* exactly $\lfloor m/2 \rfloor$ are in the upper half-plane, by setting $u = i\sqrt{t_* - t}$. For odd m the single central branch stays real, since all the other branches are accounted for in conjugate pairs. Real collision points in a compact space–time rectangle are therefore isolated and finite: an accumulation point would itself be a collision by continuity and would contradict this local description. At times $t > A^2/2$ there can be no real collision because there are only real zeros at all nearby earlier times.

Enclose an interior real collision in a small space–time cylinder with a spatial disk centered on the real axis, with root-free lateral boundary and simple zeros on the two time faces. Such cylinders can be taken arbitrarily small. Uniform convergence and Rouché's theorem preserve the counts of upper and real zeros of F on both faces for all large n; the disks about simple real zeros are taken symmetric about the axis. The upper count for F_n on such a cylinder changes only at real collisions. Its total decrease is therefore exactly $\lfloor m/2 \rfloor$, even if the approximating collision splits into several events. At an initial multiple real zero use instead a half-cylinder starting at time zero. The canonical truncations have precisely the same initial zero there for large n. Its whole cluster starts real, so no approximant can lose an upper zero inside that cylinder. Collisions at time zero are thus excluded. On a compact set outside the chosen neighborhoods, simultaneous convergence of F_n and F_n' rules out further real collisions. These facts imply convergence against continuous test functions of compact spatial support to the measure ν which assigns $\lfloor m/2 \rfloor$ at each real collision of F in $(0, T]$, projected onto the real axis. In particular local masses cannot accumulate at either time endpoint.

The tail estimate passes to ν by exhaustion with compactly supported functions. Hence $\int h_L\, d\nu_n \to \int h_L\, d\nu$. We can now pass to the limit in (3.5) to obtain

$$(1 - \eta_L)N_L^f(0) \leq \int h_L\, d\nu \leq (1 + \eta_L)N_L^f(0).$$

As $L \to \infty$ the integral increases to $\nu(\mathbb{R})$. For finite M, $N_L^f(0) \to M$ term by term. For infinite M the sum tends to infinity by retaining any fixed finite subset of upper zeros. This proves (1.2) for $a = 0$. In particular, $M > 0$ forces an actual finite real collision at a time $0 < t_* \leq A^2/2$. This proves Theorem 1.1, since a function of order less than two has no quadratic Gaussian factor in (1.1) and strict positivity forbids such a collision.

For $a > 0$ put $d_t = 1 - 4at$. Gaussian conjugation gives

$$F(t, z) = d_t^{-1/2} e^{-az^2/d_t} (e^{-(t/d_t)D^2} g)(z/d_t), \qquad 0 \leq t < 1/(4a). \tag{4.1}$$

For instance both sides are analytic near $t = 0$, satisfy $F_t = -F_{zz}$ by differentiation, and have the same initial value and hence the same time Taylor expansion. The identity extends

throughout the indicated interval by analytic continuation. The Gaussian integral for the left side is locally convergent there by the minimal growth of g. The time t/d_t runs from zero to infinity. Real rescaling and a nonzero Gaussian preserve the zeros and their multiplicities, and $t/d_t = A^2/2$ corresponds to $t = A^2/(2+4aA^2)$. This completes Theorem 1.2. Sharpness follows from $e^{-sD^2}(z^2+A^2) = z^2+A^2-2s$ and (4.1).

5 A weak test at rational times and positions

Set

$$\Delta_f(t,x) = L(F(t,\cdot))(x)\, L(F_z(t,\cdot))(x).$$

At a real collision of order m at (s,x_0), write $u = \partial_z^m F(s,x_0)$. If $m = 2r$, time Taylor expansion at $s-h$ for $h \downarrow 0$ gives leading terms $uh^r/r!$ and $uh^{r-1}/(r-1)!$ for F and F_{zz} at x_0, while $F_z = O(h^r)$. The leading term of $L(F)$ is negative. The leading term of $L(F_z)$ is the positive square of the leading term of F_{zz}. For $m = 2r+1$ apply the even-order calculation to F_z; the leading term of $L(F)$ is then $u^2h^{2r}/(r!)^2$. Thus in both cases

$$\Delta_f(s-h,x_0) \sim \begin{cases} -u^4h^{2m-3}/[r!(r-1)!^3], & m = 2r, \\ -u^4h^{2m-3}/[r!^3(r-1)!], & m = 2r+1. \end{cases} \tag{5.1}$$

In particular every collision gives an *open* set of negative values immediately before its time. The first Laguerre form alone can miss an odd collision: for $f(z) = z^3+z$ one has $F(t,z) = z^3+(1-6t)z$ and $L(F) = 3x^4+(1-6t)^2 \geq 0$ at all times.

Corollary 5.1. *For f in (1.1), all its zeros are real if and only if $\Delta_f(t,x) \geq 0$ for all positive rational $t < 1/(4a)$ and rational x (all positive rational t if $a = 0$).*

Proof. A nonreal initial zero produces a collision by Theorem 1.2. Equation (5.1) and continuity give a rational witness. Conversely if f has only real zeros, for $a = 0$ each heat-evolved truncation is a real linear exponential times a real-rooted polynomial and so belongs to the Laguerre–Pólya class. Compact limits and differentiation preserve this class and the first Laguerre inequality holds there. For $a > 0$ the same conclusion follows from (4.1), since multiplying by a real Gaussian and real rescaling also preserve the class. □

6 What the Riemann specialization requires

Let $H_t(z) = \int_0^\infty e^{tv^2}\Phi(v)\cos(zv)\,dv$ in the standard normalization of [7, 6], so that $H_0(z) = \Xi(z/2)/8$. The Newman constant Λ is the threshold for reality of all the zeros. Rodgers and Tao proved $\Lambda \geq 0$ and Platt and Trudgian proved $\Lambda \leq 1/5$ [7, 8]. Write $\Delta(t,x) = L(H_t)(x)L(H_t')(x)$.

Corollary 6.1. *In this normalization,*

$$\Lambda = \sup\big(\{0\} \cup \{t \in \mathbb{Q} : 0 < t < 1/5,\ \Delta(t,x) < 0 \text{ for some } x \in \mathbb{Q}\}\big).$$

In particular RH is equivalent to nonnegativity in this rational two-variable test.

Proof. For $t \geq \Lambda$, H_t and H_t' belong to the Laguerre–Pólya class and both factors are nonnegative. If $\Lambda > 0$, take any $0 < a < \Lambda$. At time a there is a nonreal zero. The function H_a has order one, is real on the real axis, and has bounded zero heights [6]. Theorem 1.2 applied to H_a, using the heat semigroup, gives a real collision at some time $s > a$. It satisfies $s \leq \Lambda$, since a later collision would have nonreal zeros immediately before it. Equation (5.1) gives an open set of negative witnesses in $a < t < s$, including rational ones. Let $a \uparrow \Lambda$. □

For the Riemann family alone the existence of a last collision at positive Λ also follows from the uniform large-height estimate in [6, Theorem 1.5]: zeros off the real axis have $|\operatorname{Re} z| < \exp(C/t)$ for $0 < t \le 1/2$ and an absolute C. A sequence of nonreal conjugate pairs as $t \uparrow \Lambda > 0$ therefore has a finite limit subsequence, giving a multiple real zero by Rouché's theorem. Earlier work already uses exclusion of positive-time collisions as an RH criterion [10]. At heat time zero, the different condition using $H_0(z + i\mu) + H_0(z - i\mu)$ for every $0 < \mu < 1$ is also an established RH criterion [4, Theorem 4.1, phase zero]. This relies on the known positive proportion of critical-line zeros, so does not require a general density-free converse.

Unconditional information inside the flow. Equation (1.2) implies that the number of nontrivial zeta zeros with $\operatorname{Im}\rho > 0$ and $\operatorname{Re}\rho > 1/2$, with multiplicity, equals

$$\sum_{0<t\le 1/5,\ x>0} \left\lfloor \frac{\operatorname{ord}_x H_t}{2} \right\rfloor,$$

where the sum runs over real multiple zeros of H_t. Indeed the change of variables is $\rho = 1/2 + iz/2$. The zeros of H_0 are symmetric about both axes; it has no zeros on the imaginary axis, and $H_t(0) > 0$ by its integral. Half of each count therefore gives the asserted symmetry reduction, also when both counts are infinite. At each fixed positive time all but finitely many zeros of H_t are real and simple [9]. Applying the same count at that time shows that only finitely many collision points can lie at later times. Thus an infinite set of positive-time collisions would have times tending to zero and spatial positions escaping every bounded interval.

We can also verify a region of the weak test without assuming RH. For $0 < t \le 1/5$ put $X(t) = \exp(C/t)$ with the absolute constant from [6, Theorem 1.5]. Then

$$\Delta(t, x) \ge 0 \quad \text{if } |x| \ge X(t) + 2.$$

To check this, write $h = H_t$. Its nonreal zeros $c \pm id$ have $|c| < X(t)$ and $0 < d \le 1$. Every nonreal zero of h' is either a zero of h or lies in the closed disk $(x - c)^2 + y^2 \le d^2$ of some such pair. Indeed, for $y > 0$ each real zero gives a positive term in $\partial_y \log|h(x + iy)|$, and each conjugate pair gives a term with the sign of $(x - c)^2 + y^2 - d^2$, by logarithmic differentiation. Outside all these disks the derivative is positive and h' cannot vanish. Thus the nonreal zeros of h' have $|\operatorname{Re} z| \le X(t) + 1$ and $|\operatorname{Im} z| \le 1$. For either $k = h$ or $k = h'$ use its order-one product on the real axis. A nonreal pair $c \pm id$ contributes $2((x - c)^2 - d^2)/((x - c)^2 + d^2)^2$ to $L(k)/k^2$, and each real zero contributes a positive reciprocal square. At a zero of k, $L(k) = (k')^2$. Both forms are therefore nonnegative on the stated range. This argument uses the high-height zero theorem and the localization in [5, Proposition 2.2]; it gives no numerical value of C.

The interior region $0 < t < 1/5$, $|x| < X(t) + 2$ still requires a sign proof. Likewise the first-order test $L(H_0) \ge 0$ on the entire real line is the separate Open Problem 4.7 in [5]. The general collision theorem establishes the Gaussian implication and the count. Neither a proof of RH nor a numerical improvement of the Newman upper bound is obtained here.

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